A COMPARATIVE ANALYSIS OF COUNTRY-LEVEL DATA: INITIAL FINDINGS AND INSIGHTS FROM THE CLEANED DATASET

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1. Executive summary

The files contain country-level indicators covering population, economic activity, health, education, internet access, environment and homicide. The original workbook has 204 rows and 38 columns. It includes 12 exact duplicate rows, leaving 192 unique country records after deduplication. The cleaned dataset used in the notebook contains 192 countries and 20 selected variables, with no missing values reported in those selected columns.

The main substantive finding is that homicide rates differ sharply across countries and regions. In the cleaned analysis, the median homicide rate is 2.5 while the mean is 6.5, showing that a relatively small number of countries with high rates pull the average upward. Central America, the Caribbean and Southern Africa have the highest regional median rates in the notebook's analysis. Several health, education and development indicators move together, but no single indicator explains homicide rates on its own.

The cleaned data is more convenient for analysis, but it should not be treated as fully verified. The notebook flags repeated median values in homicide and CO2 emissions, and a duplicated ISO code for Namibia. These issues matter particularly if the dataset is used for mapping, ranking countries or building a predictive model.

2. What the datasets contain

Dataset / stage	Size	What it contains
Original Dataset	204 rows × 38 columns	A broad collection of country indicators, including economic, demographic, health, education, employment, environmental and tourism measures.
Cleaned dataset	192 rows × 20 columns	A selected set of country identifiers and indicators used for the analysis, including homicide rate, population, life expectancy, GDP per capita, internet use, school enrollment, CO2 emissions and forested area.

The original workbook has one row per country in its intended structure, but repeated rows mean that 204 rows do not represent 204 distinct countries. The notebook removes exact duplicate rows and reports 192 unique countries. The cleaned analysis is therefore based on the deduplicated set, rather than the original row count.

3. Initial findings from the uncleaned dataset

3.1 Structure and data completeness

The original dataset contains 204 rows, 38 columns and 192 distinct country names. It has 12 exact duplicate rows, 12 repeated country-name entries and 12 repeated ISO-code entries. There are 305 blank cells across the workbook. Missingness is uneven: some indicators are almost complete, while others are absent for a substantial number of countries.

Column	Missing values	Share of rows
co2_emissions	59	28.9%
post_secondary_enrollment_female	33	16.2%
post_secondary_enrollment_male	33	16.2%
homicide_rate	23	11.3%
refugees	14	6.9%

Column	Missing values	Share of rows
secondary_school_enrollment_male	14	6.9%
secondary_school_enrollment_female	14	6.9%
employment_services	11	5.4%
employment_industry	11	5.4%
employment_agriculture	11	5.4%

CO2 emissions has the largest gap: 59 of 204 rows (28.9%) are blank. Female and male post-secondary enrollment each have 33 missing values (16.2%), while homicide rate is missing for 23 rows (11.3%). These gaps are important because missingness may not be random; for example, some countries may have less complete reporting for particular indicators.

3.2 Broad numerical picture

Indicator	Available rows	Median	Mean	Minimum	Maximum
Population (thousands, as stored)	204	9,227.50	39,678.04	1	1,439,324
Population density	204	88.60	335.17	2.10	26,373.80
Male life expectancy	198	70.65	69.82	50.40	81.70
Female life expectancy	198	76.70	74.65	54.80	87.50
GDP per capita	203	5,930.10	15,888.43	99.60	185,835.00
Homicide rate	181	2.50	6.83	0.00	52.00
Internet users	202	58.95	54.72	0.00	99.70
CO2 emissions	145	31.30	224.64	0.60	9,257.90
Unemployment	194	5.40	7.30	0.10	30.60
Infant mortality	196	14.35	21.94	1.30	81.90
Fertility	199	2.30	2.74	1.10	7.00

The ranges are wide for several indicators. Population and population density vary greatly between countries, and GDP per capita ranges from about 99.6 to 185,835 in the units supplied. Homicide rates range from 0 to 52 among the 181 countries with a recorded value. Such wide ranges suggest that averages alone will not describe a typical country very well; medians and regional breakdowns are useful complements.

3.3 What stands out before cleaning

- The original data needs deduplication before country counts, averages or correlations are calculated. Otherwise, countries represented by duplicate rows may receive extra weight.
- The largest missing-data problem is in CO2 emissions, followed by post-secondary enrollment and homicide rate. Analyses using these fields will have fewer observed countries unless missing values are imputed.
- The dataset covers 22 regions, but region sizes differ. Regional comparisons should show the number of countries behind each estimate, especially for small regions.
- The workbook includes a mixture of identifiers, categorical fields and numeric indicators. Currency, capital, country name, region and ISO code should be treated differently from measured numeric variables.
- These are initial findings from the workbook as supplied. They are descriptive only and do not imply that any relationship is causal.

4. Findings from the cleaned dataset

4.1 Homicide rates are unevenly distributed

The cleaned notebook reports 192 countries. Homicide rates have a mean of 6.5 and a median of 2.5. The mean being substantially higher than the median is a sign of a right-skewed distribution: most countries have comparatively low rates, while a smaller group has much higher rates.

Homicide-rate category	Countries	Share of 192
Low (≤ 2)	68	35.4%
Moderate (> 2 to 10)	91	47.4%
High (> 10 to 20)	14	7.3%
Very high (> 20)	19	9.9%

Together, 159 of 192 countries (82.8%) fall in the low or moderate categories, with a homicide rate of 10 or below. The maximum recorded rate is 52.0, for El Salvador. The notebook identifies 23 countries above the statistical upper-outlier fence of 14.43. This fence is a method for flagging unusually high values, not a judgement that the records are incorrect.

The notebook also notes that 26 countries share a homicide rate of exactly 2.5. Because 2.5 is the dataset median, this concentration is consistent with median imputation. That is plausible, but the value pattern alone does not prove how each value was created. If these values were imputed, they should not be interpreted as measured country-specific rates.

4.2 Regional differences are pronounced

Region	Median homicide rate reported	Interpretation
Central America	25.8	Highest regional median in the analysis. Second-highest
Caribbean	21.4	regional median.
Southern Africa	17.7	Third-highest regional median.
South America	7.5	Elevated relative to most other regions, but below the top three.
Eastern Africa	4.9	Below the first four regions, but above many others.
Southern Europe	0.8	Among the lowest regional medians.
Western Europe	0.9	Among the lowest regional medians.
Eastern Asia	0.6	Lowest regional median among the examples reported.

The difference between regions is more pronounced than a single global average suggests. The notebook reports that all other regions have median rates of 3.4 or lower, apart from the higher-rate regions listed above. These comparisons are descriptive; differences in reporting practices, data coverage, population characteristics and other factors may also influence the observed values.

4.3 Economic and social indicators

GDP per capita is strongly right-skewed in the cleaned dataset: the notebook reports a mean of about US$15,500 and a median of about US$5,900. A small number of high-income countries pull the mean upward, so the median is a better summary of the middle country in this dataset.

Population, population density and CO2 emissions also have long right tails. Unemployment and infant mortality are right-skewed as well, while life expectancy and school enrollment tend to cluster toward the higher end of their observed ranges. These patterns point to substantial differences in country size and development conditions; they also explain why comparisons based only on means can be misleading.

The notebook's strongest indicator relationships include female and male secondary-school enrollment (Spearman/Pearson-style correlation reported as 0.97), male and female life expectancy (0.96), female life expectancy and infant mortality (-0.94), and population and CO2 emissions (0.81). These patterns are intuitive associations, but they also mean that several variables carry overlapping information. A model using all of them may face multicollinearity.

4.4 Homicide and other indicators

The cleaned analysis finds that homicide rate has mostly weak-to-moderate relationships with individual indicators. Spearman correlations are negative for male life expectancy (-0.43), female life expectancy (-0.39), internet use (-0.37) and GDP per capita (-0.33). They are positive for infant mortality (+0.41), fertility (+0.34) and unemployment (+0.19).

These figures describe rank-based associations, not cause and effect. The difference between Pearson and Spearman results for some variables suggests that relationships may not be straight-line patterns and may be affected by extreme values. It would be misleading to conclude that changing any one of these indicators would directly change homicide rates.

5. Cleaning and data-quality notes

- Duplicate rows: the notebook removes 12 exact duplicates, reducing the dataset from 204 rows to 192 unique country records.
- ISO code inconsistency: Namibia is assigned the code AD in the notebook's data, which is also assigned to Andorra. The notebook notes that Namibia's expected code is NA. This should be corrected before map-based analysis.
- Repeated median values: CO2 emissions has 31.3 repeated for 57 countries in the cleaned dataset; homicide rate has 2.5 repeated for 26 countries. The notebook interprets these patterns as consistent with median imputation and checks the impact of excluding the repeated homicide values.
- Sensitivity check: after excluding the 26 countries with homicide rate exactly 2.5, the notebook reports that the leading regional medians and the 23 flagged high-rate countries remain unchanged. Some rank correlations become slightly stronger. This suggests that the main patterns are not solely created by the repeated median values.
- Selected variables: the cleaned notebook uses 20 columns rather than all 38 in the original workbook. Findings from the cleaned analysis should therefore be understood as applying to this selected subset of indicators.

6. What the findings mean overall

Taken together, the data presents a broad picture of differences in living conditions and social outcomes between countries. Population size, economic resources, health outcomes, education and access to the internet vary considerably. Several of these indicators are closely related to one another, which is expected when they reflect wider differences in development.

Homicide is the most unevenly distributed outcome examined in the notebook. The high regional medians in Central America, the Caribbean and Southern Africa stand out, while most countries fall at or below a rate of 10. Individual economic or social measures show some association with homicide, but the relationships are not strong enough to explain the pattern by themselves. Regional context and other unmeasured factors are likely to matter.

For practical use, the cleaned dataset is a reasonable starting point for descriptive analysis and for developing hypotheses about possible crime hotspots. It is not yet sufficient on its own to establish causes or to make reliable predictions. Before a mapping or prediction project, the ISO codes, imputation history, indicator definitions, units and reference years should be verified.

7. Limitations and recommended next steps

Confirm the source and reference year for every indicator. Country-level figures may come from different years, so they may not represent a single consistent snapshot.

Document the cleaning process, especially which missing values were imputed, which method was used and whether the imputed values can be distinguished from observed values.

Correct and validate country identifiers against a trusted country-code list before using maps or joining external datasets.

For every regional comparison, report the number of countries with available homicide data as well as the median. Small groups can produce unstable estimates.

Keep outlier countries in descriptive reporting unless there is evidence of an error. Investigate them separately rather than automatically deleting them.

For predictive modelling, establish a clear target definition, avoid data leakage, test the model on held-out data and report uncertainty. Correlation alone is not evidence of predictive performance.

Consider comparing results with and without imputed records, and document any changes. The notebook's existing sensitivity check is a useful first step.

8. Conclusion

The original dataset is rich in indicators but needs care before analysis because of duplicate rows and uneven missingness. The cleaned dataset makes the country comparisons easier to interpret and reveals clear differences in homicide rates, regional patterns and development indicators. The most important takeaway is not that one variable explains crime, but that country outcomes vary widely and that the observed relationships are complex. The findings are useful as an initial descriptive report, provided the remaining data-quality limitations are kept in view.